MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3750 TAILING & CIL WATER PROCESS STORAGE
DESIGN CALCULATION OF 3750-TNK-001

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
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QA	:	Quality Assurance
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1. INTENT

This document is to Prepare for design of **TAILING & CIL WATER PROCESS STORAGE (3750-TNK-001)** Tank Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in excel and STAAD PRO. For member strength calculation force values are directly taken from load data provided and output of STAAD analysis.

4. DESIGN REQUIREMENTS

4.1 Design Loads

All Design Loads are indicated in Doc. No. MD-216-3750-EG-ST-GAD- 1001 attached in Attachment 2. Loads are also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

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5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations for Aboveground Tanks

Foundation shall be Raft Foundation with a Concrete Ringwall.

5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Wall & Raft	50

5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m^2 at 1.5m depth below EGL. (refer Table 5 - BH-14 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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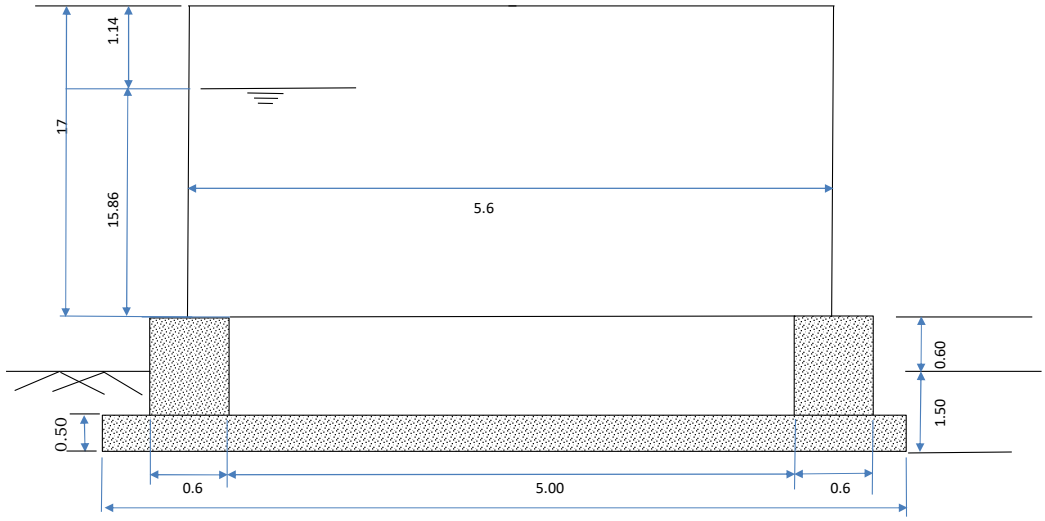
Foundation for 3750-TNK-001

Tank Load Data

DEAD WEIGHT OF TANK	270	kN
ROOF PIPE LIQUID LOAD	53	kN
SHELL PIPE LIQUID LOAD	53	kNm
TANK LIQUID LOAD (OPERATING)	3830	kN
TANK LIQUID LOAD (TEST)	4108	kN
LIVE LOAD	84	kN
WIND LOAD - HORIZONTAL	54	kN
WIND LOAD - VERTICAL	-27	kN
SEISMIC LOAD	65	kN

	Loading Type		
Weight, V	Empty (W)	270	kN
	Operating Condition (Wo)	4206	kN
	Test Condition (Wt)	4484	kN

Summary of Tank Specification



Diameter of Tank, D	=	5.6 m	
Height of Tank, H	=	17 m	
Height of Tank Fluid, H_L	=	15.86 m	
Inner Diameter of Concrete Ring Wall, ID	=	5 m	
Outer Diameter of Concrete Ring Wall, OD	=	6.2 m	
Height of Concrete Ring Wall, h	=	2.10 m	
Thickness of Concrete Ring Wall, t	=	0.6 m	
Thickness of Concrete Raft, T	=	0.5 m	
Top of Foundation	=	0.6 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L	=	10 kN/m^3	
Unit Weight of Soil, γ_s	=	20 kN/m^3	
Unit Weight of Concrete, γ_c	=	24 kN/m^3	

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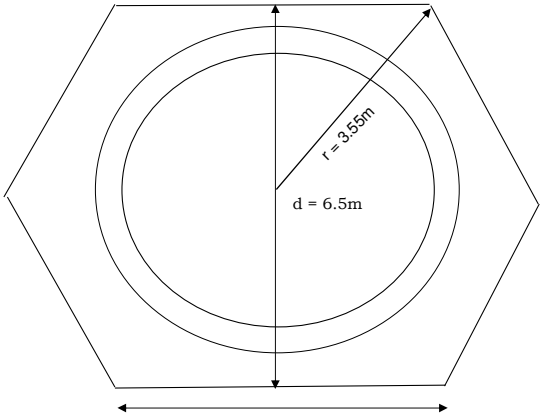
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Wind Load & Seismic Load

Total Base Shear due to wind load = 54 kN

Total Base Shear due to Seismic load = 65 kN

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	54.00	560.00
	Operating	54.00	560.00
Seismic Loads	Operating	65.00	499.00



Foundation Weight

a) Ring Wall Weight = 405.34 kN

b) Inside Soil Weight = 628.32 kN

c) Outside Soil Weight = 51.01 kN

d) Raft Weight = 392.90 kN

Area of Raft

Area of Raft (Hexagonal), $A = 2.598 * r^2$ = 32.74 m^2

Property of Hexagon

Radious of outer circle, r = 3.55 m

Dist. of 2 opposite side, d = 6.5 m

Sectional Modulus of Raft

Sectional modulus (Hexagonal), $Ze = 0.0601*d^4 / r$ = 30.22 m^3

Basic Load Cases

1. Tank Dead Load (D)
2. Live Load (L)
3. Operating liquid Load (F1)
4. Test liquid Load (F2)
5. Wind Load (W)
6. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4.1)

Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Operating	LC101	D + F1	4206.00	0.00	0.00
Empty	LC102	0.6 D + W	162.00	54.00	560.00
Operating	LC103	D + F1 + W	4179.00	54.00	560.00
	LC104	D + F1+ 0.7 E	4206.00	45.50	349.30
	LC105	0.6D + 0.7 E	2523.60	45.50	349.30
Test	LC106	D + F2	4484.00	0.00	0.00
	LC107	D + F2 + L	4568.00	0.00	0.00
	LC108	D + F2 + W	4457.00	54.00	560.00
	LC109	D + F2 + 0.7 E	4484.00	45.50	349.30
Operating	LC110	D + F1 + L	4290.00	0.00	0.00

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Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 450 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z	Qmax= V/A - M/Z	Qall/ Qmax	Soil Bearing Check
	V (kN)	M(kNm)	kN/m^2	kN/m^2		
LC101	5683.57	0.00	173.59	173.59	2.59	OK
LC102	1639.57	673.40	72.36	27.79	6.22	OK
LC103	5656.57	673.40	195.05	150.48	2.31	OK
LC104	5683.57	444.85	188.31	158.87	2.39	OK
LC105	4001.17	444.85	136.93	107.49	3.29	OK
LC106	5961.57	0.00	182.08	182.08	2.47	OK
LC107	6045.57	0.00	184.65	184.65	2.44	OK
LC108	5934.57	673.40	203.54	158.97	2.21	OK
LC109	5961.57	444.85	196.80	167.36	2.29	OK
LC110	5767.57	0.00	176.16	176.16	2.55	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	5683.57	0.3	1705.07	0.00	-	-
LC102	1639.57	0.3	491.87	54.00	9.1	OK
LC103	5656.57	0.3	1696.97	54.00	31.4	OK
LC104	5683.57	0.3	1705.07	45.50	37.5	OK
LC105	4001.17	0.3	1200.35	45.50	26.4	OK
LC106	5961.57	0.3	1788.47	0.00	-	-
LC107	6045.57	0.3	1813.67	0.00	-	-
LC108	5934.57	0.3	1780.37	54.00	33.0	OK
LC109	5961.57	0.3	1788.47	45.50	39.3	OK
LC110	5767.57	0.3	1730.27	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Edge Dis.(d/2)	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	5683.57	3.25	18471.59	0.00	-	-
LC102	1639.57	3.25	5328.59	673.40	7.9	OK
LC103	5656.57	3.25	18383.84	673.40	27.3	OK
LC104	5683.57	3.25	18471.59	444.85	41.5	OK
LC105	4001.17	3.25	13003.79	444.85	29.2	OK
LC106	5961.57	3.25	19375.09	0.00	-	-
LC107	6045.57	3.25	19648.09	0.00	-	-
LC108	5934.57	3.25	19287.34	673.40	28.6	OK
LC109	5961.57	3.25	19375.09	444.85	43.6	OK
LC110	5767.57	3.25	18744.59	0.00	-	-

Ultimate Strength Design

Load Combination (As per SBC-304 , Table 5.3.1)

Load Conditions	Load Case	Load Combination	Load at bottom of tank Inner Area			
			V (kN)	M (kNm)	A (m^2)	Z (m^3)
Operating	LC201	1.4 (D + F1)	5888.40	0.00	24.63	17.24
	LC202	1.4 (D + F1) + 1.7 L	6031.20	0.00	24.63	17.24
Empty	LC203	0.9 D + 1.6 W	199.80	896.00	24.63	17.24
Operating	LC204	1.2(D + F1) + 1.6W + L	5088.00	896.00	24.63	17.24
	LC205	1.2 (D + F1) + E + L	5131.20	499.00	24.63	17.24
	LC206	0.9 (D + F1) + E	3785.40	499.00	24.63	17.24
Test	LC207	1.4 (D + F2)	6277.60	0.00	24.63	17.24
	LC208	1.4 (D + F2) + 1.7 L	6420.40	0.00	24.63	17.24
	LC209	1.2(D + F2) + 1.6W + L	5421.60	896.00	24.63	17.24
	LC210	1.2 (D + F2) + E + L	5464.80	499.00	24.63	17.24

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Staad input calculation.

Total Number of Noads in Wall Top = 24

S/W of tank (Empty Condition) on each Nodes= 11.25 kN

Pressure due to Sand fill on inside Raft= 32.0 kN/m^2

Pressure due to soil fill on outside Raft= 20.0 kN/m^3

Wind Base Shear of tank on each Nodes= 2.250 kN

Seismic Base Shear of tank on each Nodes= 2.708 kN

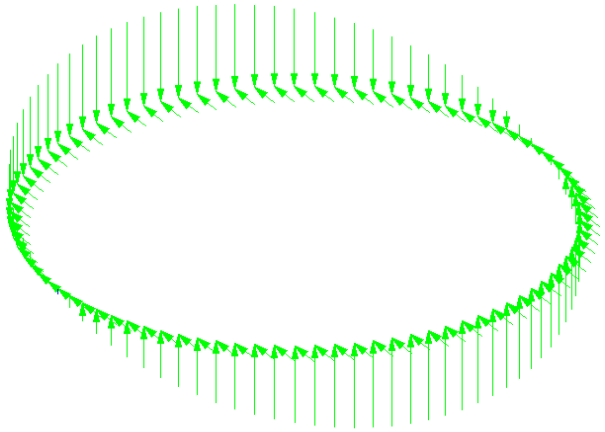
Sl No.	Node No.	Dist. R(m)	R^2	Seismic V(kN)	Wind V(kN)
1	3042	0.00	0.00	0.0	0.0
2	3043	0.72	0.53	3.8	4.3
3	3044	1.40	1.96	7.4	8.3
4	3045	1.98	3.92	10.5	11.8
5	3046	2.43	5.88	12.9	14.4
6	3047	2.70	7.32	14.3	16.1
7	3048	2.80	7.84	14.8	16.7
8	3049	2.70	7.32	14.3	16.1
9	3050	2.43	5.88	12.9	14.4
10	3051	1.98	3.92	10.5	11.8
11	3052	1.40	1.96	7.4	8.3
12	3053	0.72	0.53	3.8	4.3
13	3054	0.00	0.00	0.0	0.0
14	3055	-0.72	0.53	-3.8	-4.3
15	3056	-1.40	1.96	-7.4	-8.3
16	3057	-1.98	3.92	-10.5	-11.8
17	3058	-2.43	5.88	-12.9	-14.4
18	3059	-2.70	7.32	-14.3	-16.1
19	3060	-2.80	7.84	-14.8	-16.7
20	3061	-2.70	7.32	-14.3	-16.1
21	3038	-1.98	3.92	-10.5	-11.8
22	3039	-2.43	5.88	-12.9	-14.4
23	3040	-1.40	1.96	-7.4	-8.3
24	3041	-0.72	0.53	-3.8	-4.3
		ΣR^2=	**94.10**		

Staad input calculation For Seismic and Wind load distribution.

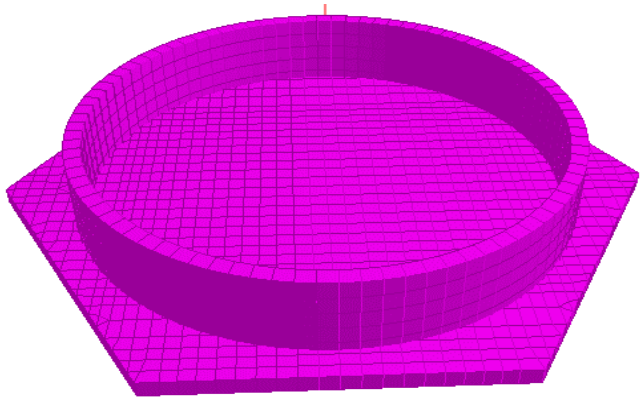
$V_{wind} = R * W_{total} / \Sigma R^2$ $\quad H_{wind} = W_{total} / 24$

$V_{Seismic} = R * E_{total} / \Sigma R^2$ $\quad H_{seismic} = E_{total} / 24$

H_{wind}=	2.250	kN
$H_{seismic}$=	2.708	kN



Staad Model in 3D view



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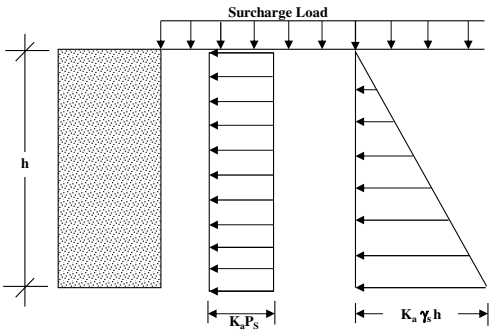
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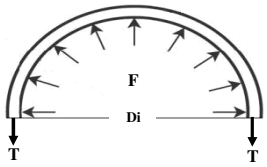
Wall Design

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m^2
LC201	5888.40	0.00	239.07
LC202	6031.20	0.00	244.87
LC203	199.80	896.00	60.08
LC204	5088.00	896.00	258.55
LC205	5131.20	499.00	237.27
LC206	3785.40	499.00	182.63
LC207	6277.60	0.00	254.88
LC208	6420.40	0.00	260.67
LC209	5421.60	896.00	272.09
LC210	5464.80	499.00	250.82



Maximum Surcharge Load, Ps = 272.09 kN/m^2



Where, K_a = coefficient of earth pressure at rest = 0.5

Pressure due to Surcharge,P1 = $K_a * P_s$ = 136.04 KN/m^2

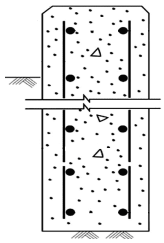
Pressure due to Soil,P2 = $1.4 * \gamma_S * (h-T) * K_a$ = 22.40 KN/m^2 for LC209 - 1.2(D + F2) + 1.6W + L

Maximum Lateral Load

F = P1 * (h-T) + 1/2 * P2 *(h -T)= 235.592 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 588.98 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(ϕ x Fy) = 2400 mm^2

Minimum As = 0.0025 x t x h = 2400 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 20 Nos. D16= 3629.168 mm^2 OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 900 mm^2 (As per API 650 , Cl. No. B.4.2.3)

STAAD output:- (Refer Attachment 1)

Wall Thickness = 600 mm

Characteristic strength of concrete, f '$_c$= 35 MPa

Yield stress of steel reinforcement, f_y= 420 MPa

Clear cover for main reinforcement bars = 50mm

Max BM (+) (M_y)=	46.533	kN m/m	(VERTICAL MOMENT IN INSIDE FACE)
Max BM (-) (M_y)=	6.816	kN m/m	(VERTICAL MOMENT IN OUTSIDE FACE)
Max BM (+) (M_x)=	8.325	kN m/m	(HORIZONTAL MOMENT IN INSIDE FACE)
Max BM (-) (M_x)=	6.033	kN m/m	(HORIZONTAL MOMENT IN OUTSIDE FACE)

Max Shear Qx = 0.023 N/mm^2

Max Shear Qy = 0.166 N/mm^2

Effective depth of wall in Vertical direction = 542 mm

Effective depth of wall in Horizontal direction = 526 mm

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Moment	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
My(+)	46.53	0.16	0.21	0.001	273.53	975.60	D16	206.09	300	175	1148.93
My(-)	6.82	0.02	0.03	0.000	39.94	975.60	D16	206.09	300	175	1148.93
Mx(+)	8.33	0.03	0.04	0.000	50.28	975.60	D16	206.09	300	175	1148.93
Mx(-)	6.03	0.02	0.03	0.000	36.43	975.60	D16	206.09	300	175	1148.93

Check against one-way shear force

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.74 N/mm^2

Shear stress at a distance 'd' from raft top , τ_v= 0.17 N/mm^2 **OK**

SUMMARY OF REINFORCEMENT IN WALL: **600 mm**

VERTICAL REIN. IN INSIDE FACE **D16 175 c/c**

VERTICAL REIN. IN OUTSIDE FACE **D16 175 c/c**

HORIZONTAL REIN. IN INSIDE FACE **D16 175 c/c (10Nos.)**

HORIZONTAL REIN. IN OUTSIDE FACE) **D16 175 c/c (10Nos.)**

Raft Design

STAAD output:- (Refer Attachment 1)

Raft Thickness = 500 mm

Max BM (+) (M_y)= 22.40 kN m/m (HORIZONTAL MOMENT IN BOTTOM FACE)

Max BM (-) (M_y)= 26.314 kN m/m (HORIZONTAL MOMENT IN TOP FACE)

Max BM (+) (M_x)= 20.55 kN m/m (LONGITUDINAL MOMENT IN BOTTOM FACE)

Max BM (-) (M_x)= 26.824 kN m/m (LONGITUDINAL MOMENT IN TOP FACE)

Max Shear Qx = 0.124 N/mm^2

Max Shear Qy = 0.143 N/mm^2

Effective depth of Raft in Vertical direction = 417 mm

Effective depth of Raft in Horizontal direction = 401 mm

Moment	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
My(+)	22.40	0.13	0.17	0.000	171.01	750.60	D16	267.87	300	175	1148.93
My(-)	26.31	0.15	0.20	0.000	201.01	750.60	D16	267.87	300	175	1148.93
Mx(+)	20.55	0.13	0.17	0.000	163.19	721.80	D16	278.56	300	175	1148.93
Mx(-)	26.82	0.17	0.22	0.001	213.16	721.80	D16	278.56	300	175	1148.93

Check against one-way shear force

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.74 N/mm^2

Shear stress at a distance 'd' from raft top , τ_v= 0.14 N/mm^2 **OK**

SUMMARY OF REINFORCEMENT IN RAFT: **500 mm**

(HORIZONTAL REIN. IN TOP FACE) **D16 175 c/c**

(HORIZONTAL REIN. IN BOTTOM FACE) **D16 175 c/c**

(LONGITUDINAL REIN. IN TOP FACE) **D16 175 c/c**

(LONGITUDINAL REIN. IN BOTTOM FACE) **D16 175 c/c**

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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ATTACHMENT 1 PLATE CENTRE STRESS SUMMARY

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Plate Center Stress Summary

			Shear		Membrane			Bending		
	Plate	L/C	Qx (N/mm^2)	Qy (N/mm^2)	Sx (N/mm^2)	Sy (N/mm^2)	Sxy (N/mm^2)	Mx (kNm/m)	My (kNm/m)	Mxy (kNm/m)
Max Qx	7222	208:1.4 (D + F	**0.124**	0.028	0.000	0.000	0.000	16.023	7.591	4.895
Min Qx	7223	208:1.4 (D + F	**-0.124**	0.028	0.000	0.000	0.000	16.037	7.626	-4.915
Max Qy	7325	208:1.4 (D + F	-0.118	**0.127**	0.000	0.000	0.000	8.340	7.154	-5.222
Min Qy	7248	208:1.4 (D + F	-0.001	**-0.143**	0.000	0.000	0.000	2.393	22.398	-0.637
Max Sx	7337	201:1.4 (D + F	0.084	-0.012	**0.000**	0.000	0.000	11.068	3.364	0.768
Min Sx	7337	201:1.4 (D + F	0.084	-0.012	**0.000**	0.000	0.000	11.068	3.364	0.768
Max Sy	7337	201:1.4 (D + F	0.084	-0.012	0.000	**0.000**	0.000	11.068	3.364	0.768
Min Sy	7337	201:1.4 (D + F	0.084	-0.012	0.000	**0.000**	0.000	11.068	3.364	0.768
Max Sxy	7337	201:1.4 (D + F	0.084	-0.012	0.000	0.000	**0.000**	11.068	3.364	0.768
Min Sxy	7337	201:1.4 (D + F	0.084	-0.012	0.000	0.000	**0.000**	11.068	3.364	0.768
Max Mx	7226	208:1.4 (D + F	-0.120	0.058	0.000	0.000	0.000	**20.554**	8.294	-8.469
Min Mx	6988	207:1.4 (D + F	-0.088	-0.028	0.000	0.000	0.000	**-26.824**	-8.201	-10.755
Max My	7251	208:1.4 (D + F	0.001	-0.143	0.000	0.000	0.000	2.392	**22.398**	0.637
Min My	6802	207:1.4 (D + F	-0.006	0.110	0.000	0.000	0.000	-3.687	**-26.314**	-1.299
Max Mxy	6984	207:1.4 (D + F	0.072	-0.045	0.000	0.000	0.000	-22.412	-11.532	**13.709**
Min Mxy	6987	207:1.4 (D + F	-0.072	-0.045	0.000	0.000	0.000	-22.412	-11.537	**-13.712**

Plate Center Stress Summary

			Shear		Membrane			Bending		
	Plate	L/C	Qx (N/mm^2)	Qy (N/mm^2)	Sx (N/mm^2)	Sy (N/mm^2)	Sxy (N/mm^2)	Mx (kNm/m)	My (kNm/m)	Mxy (kNm/m)
Max Qx	6496	208:1.4 (D + F	**0.023**	-0.010	0.597	-0.059	-0.015	-0.462	2.678	2.298
Min Qx	6491	208:1.4 (D + F	**-0.023**	-0.010	0.597	-0.059	0.015	-0.469	2.678	-2.297
Max Qy	6534	207:1.4 (D + F	0.000	**0.166**	0.021	-0.130	-0.005	8.325	46.533	2.091
Min Qy	6458	203:0.9 D + 1.6	-0.001	**-0.011**	0.036	-0.117	-0.000	-0.749	-3.337	-0.051
Max Sx	6494	208:1.4 (D + F	0.007	-0.009	**0.663**	-0.062	-0.004	7.906	3.082	0.811
Min Sx	6553	203:0.9 D + 1.6	0.002	-0.007	**0.003**	-0.127	-0.001	-1.151	-6.343	0.308
Max Sy	6494	203:0.9 D + 1.6	0.001	-0.007	0.007	**0.032**	0.001	0.500	-1.410	0.144
Min Sy	6555	209:1.2 (D + F	-0.002	0.140	0.011	**-0.171**	0.002	7.026	39.384	-0.081
Max Sxy	6539	208:1.4 (D + F	-0.015	0.138	0.036	-0.101	**0.057**	5.445	33.382	-6.024
Min Sxy	6544	208:1.4 (D + F	0.015	0.138	0.036	-0.101	**-0.057**	5.445	33.378	6.025
Max Mx	6534	207:1.4 (D + F	0.000	0.166	0.021	-0.130	-0.005	**8.325**	46.533	2.091
Min Mx	6489	208:1.4 (D + F	-0.001	-0.010	0.551	-0.057	-0.003	**-6.033**	2.454	-0.817
Max My	6534	207:1.4 (D + F	0.000	0.166	0.021	-0.130	-0.005	8.325	**46.533**	2.091
Min My	6546	203:0.9 D + 1.6	-0.000	-0.005	0.014	-0.031	0.016	-1.246	**-6.816**	0.445
Max Mxy	6544	208:1.4 (D + F	0.015	0.138	0.036	-0.101	-0.057	5.445	33.378	**6.025**
Min Mxy	6539	208:1.4 (D + F	-0.015	0.138	0.036	-0.101	0.057	5.445	33.382	**-6.024**

ATTACHMENT 2 LOAD DATA DRAWINGS

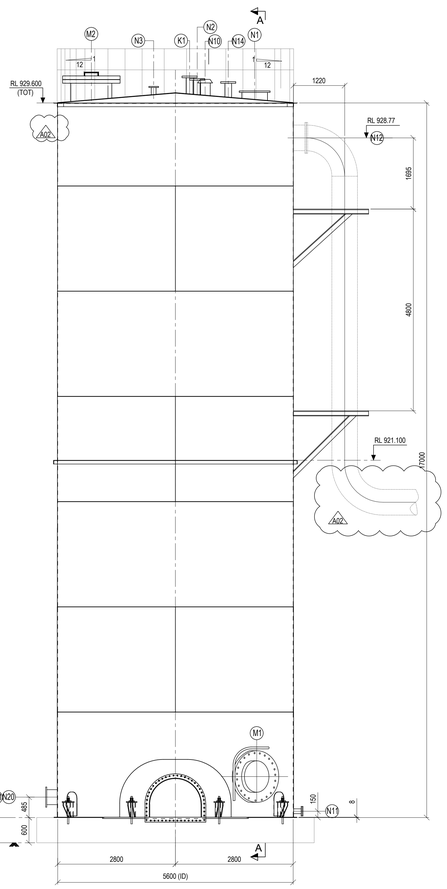
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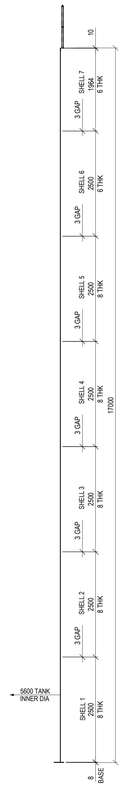
DRAWING NUMBER: MD-216-3750-EG-ST-GAD-1001 REV A02

MATERIAL SPECIFICATION	
SHELL, BOTTOM, ANNULAR, ROOF	EN 10025 GR.S275 J0
COD	EN 10025 GR.S275 J0
MANWAY NECKS/ MANWAY FLANGES	EN 10025 GR.S275 J0
NOZZLE NECKS (PIPES) (600 NB & BELOW)	ASTM A53 Gr.B/ ASTM A106 Gr.B
NOZZLE NECKS (PIPES) (> 600 NB)	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
NOZZLE FLANGES/ BLIND FLANGE	SA 105
GASKETS FOR NOZZLES	NON METALLIC FLAT RING 1.5mm/ 3mm THK, PTFE, CNAF A02
GASKETS FOR MANWAY/ COD	3mm THICK COMPRESSED NON ASBESTOS SHEETS
BOLTS/ NUTS FOR NOZZLE	SA 193 Gr. B7 / 194 Gr. 2H
BOLTS/ NUTS FOR MANWAY	SA 193 Gr. B7 / 194 Gr. 2H
STRUCTURAL BOLTS	ASTM A325M
ANCHOR CHAIR	EN 10025 GR.S275 J0
STRUCTURAL(ROLLED SECTION)-STAIRWAY / PLATFORM / ROOF SUPPORTING STRUCTURE	EN 10025 GR.S275 JR
HANDRAIL PIPE / HANDRAIL POST	ASTM A53 TYPE E
PIPE FITTINGS	SA 234 Gr. WPB
REINFORCEMENT PAD/GUSSET/ EXTERNAL SUPPORT ETC.	EN 10025 GR.S275 J0
EARTHING LUGS	EN 10025 GR.S275 J0
ANCHOR BOLTS	ASTM F1554 GR.36
CURB ANGLE	EN 10025 GR.S275 JR
NAME PLATE	STAINLESS STEEL(SS 316)

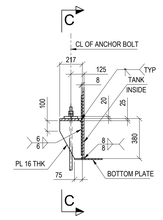
FOUNDATION LOAD DATA:					
BASIC LOAD CASE	VERTICAL COMPRESSION	VERTICAL UPLIFT	HORIZONTAL SHEAR	OVERTURNING MOMENT (ABOUT SHELL BOTTOM JOINT)	SLAB MOMENT
	(KN)	(KN)	(KN)	(KN-M)	(KN-M)
DEAD LOAD (NOMINAL THICKNESS)	270	–	–	–	–
DEAD LOAD (CORRODED THICKNESS)	221	–	–	–	–
ROOF PIPE LIQUID LOAD	53	–	–	–	–
SHELL PIPE LIQUID LOAD	53	–	–	–	–
LIVE LOAD	84	–	–	–	–
TANK LIQUID LOAD (OPERATING)	3830	–	–	–	–
TANK LIQUID LOAD (TEST)	4108	–	–	–	–
WIND LOAD ON SHELL	–	–	54	485	–
WIND LOAD ON ROOF	–	27	–	75	–
SEISMIC LOAD	134	134	65	499	553



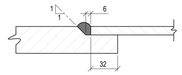
TANK ELEVATION
(SCALE 1:50)



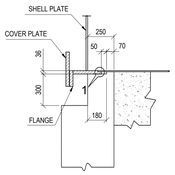
SECTION A - A
(SCALE - 1: 50)



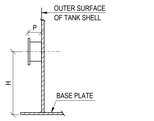
TYP DETAIL OF ANCHOR CHAIR
SCALE- 1:20



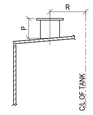
DETAIL - 1
(SCALE - 1: 10)



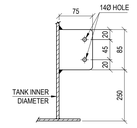
SECTION G-G
SCALE- 1:20



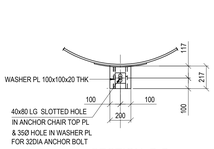
SHELL NOZZLES / FITTINGS A02



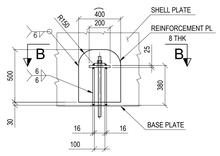
ROOF NOZZLES / FITTINGS A02



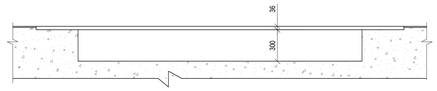
DETAIL OF EARTHING LUGS
(QUANTITY : 2 Nos.) A02



SECTION B-B
SCALE- 1:20



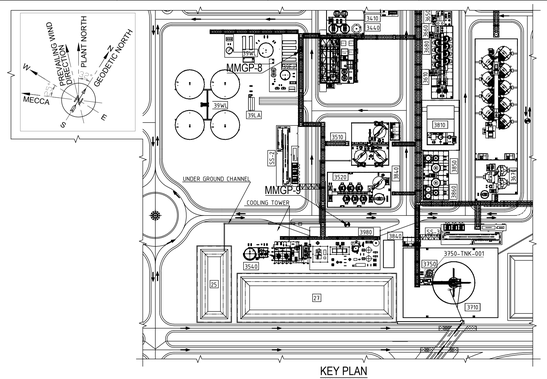
SECTION C-C
SCALE- 1:20



SECTION F - F
(SCALE - 1: 20)

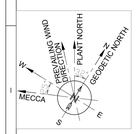
NOZZLE TABLE												
NOZ	SERVICE	SIZE (DN)		RATING #	TYPE	ELEV "H"	PROJ "P"	RADIUS "R"	ORIENTATION	LOCATION	RF PAD DIA ★	RF PAD THK. ★	REMARK
N1	INLET-THICKENER OVERFLOW	600	Sch XS	CL 150	RFSO	-	300	2300	125°	ROOF	1150	6	
N2	INLET-RAW WATER	150	Sch XS	CL 150	RFSO	-	200	2270	180°	ROOF	400	6	
N3	RE-CIRCULATION LINE	100	Sch XS	CL 150	RFSO	- A02	200	2268	245°	ROOF	305	6	
N10	TANK VENT	400x400 (TBC)	-	RECTANGULAR WITH COVER ARRANGEMENT		-	300	1575 A02	335°	ROOF	-	-	
N11	DRAIN	100	Sch XS	CL 150	RFSO	0.150	300 A02		305°	SHELL	-	-	
N12	OVERFLOW	600	Sch XS	CL 150	RFSO	16.170	300	-	0°	SHELL	1255	6	
N13	CLEAN OUT DOOR (COD)-hxb	900x1200	-	API 650		-	-		335°	SHELL	3000x1372 A02	8	
N14	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	-	400	1700 A02	65°	ROOF	275	6	
N20	OUTLET-CIL PROCESS WATER	350	Sch XS	CL 150	RFSO	0.485	250		290°	SHELL	750	8	
N21	OUTLET-CIL PROCESS WATER	350	Sch XS	CL 150	RFSO	0.485	250		250°	SHELL	750	8	
K1	LEVEL TRANSMITTER	100	Sch XS	CL 150	RFSO	-	400	1700 A02	160°	ROOF	305	6	
M1	MANHOLE C/W DAVIT	750	-	API 650		0.925	175		68°	SHELL	1545	8	
M2	MANHOLE C/W HINGED COVER	600	-			-	500	2000	275°	ROOF	1150	6	

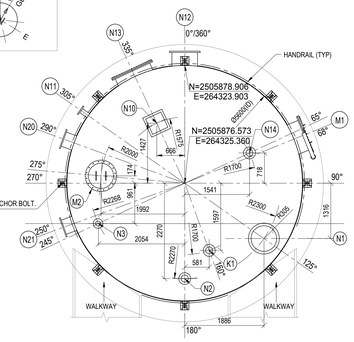
★ REFER NOTES NO. 12



KEY PLAN
(SCALE-1:1500)

DESIGN DATA			
TANK DETAILS	LOCATION		MANSOURAH - MASSARAH GOLD PROJECT
	TANK NAME		CIL PROCESS WATER TANK
	EQUIPMENT TAG No.		3750-TNK-001
	TANK DIMENSIONS		INSIDE DIAMETER : 5600mm, HEIGHT: 17000mm
	QUANTITY (NUMBER)		1
	ROOF TYPE & SLOPE		SLOPED CONE ROOF (SLOPE 1 IN 12) CONICAL APEX UP
	BOTTOM TYPE & SLOPE		SLOPED BOTTOM (SLOPE 1 IN 120) CONICAL APEX UP
	FIELD ERECTED/SHOP FABRICATED		FIELD ERECTED
	STORED FLUID		CIL PROCESS WATER
DESIGN CODE	DESIGN CODE		API 650-TWELFTH EDITION, MARCH 2013, ADDENDUM 3, AUGUST 2018
OPERATING DATA	OPERATING TEMPERATURE		36° C
	NET WORKING CAPACITY		350 m^3
	MAXIMUM CAPACITY		390 m^3 (TBC)
	OPERATING LEVEL	MINIMUM	1340 mm
		MAXIMUM	15850 mm
	OPERATING DENSITY OF CONTENT		1.0 t/m^3
MECHANICAL & PROCESS DATA	DESIGN TEMPERATURE		50° C
	DESIGN PRESSURE (INTERNAL/EXTERNAL)		ATM/ATM
	TEST PRESSURE		ATM
	HYDRO STATIC TEST PRESSURE		FULL OF WATER
	DESIGN DENSITY OF CONTENT		1.0 t/m^3
	TESTING		AS PER API 650
	INSPECTION		AS PER API 650
	DENSITY OF SCALE		N/A
	DESIGN SCALE THICKNESS	TANK SHELL/ROOF	N/A
	DESIGN FILL RATE		362 m/hr (TBC)
	DESIGN EMPTY RATE		362 m/hr (TBC)
	CORROSION ALLOWANCE	SHELL	1.6 mm
		ROOF PLATE	0 mm
		BOTTOM PLATE / ANNULAR PLATE	1.6 mm
		ANCHOR BOLT	0 mm
		STRUCTURAL	1.6 mm
WIND LOAD DATA	BASIC WIND SPEED		43 m/sec. (155 Km/hr.)
	SITE EXPOSURE		C
	VELOCITY PRESSURE EXPOSURE COEFFICIENT		K_z= AS PER ELEVATION
	TOPOGRAPHIC FACTOR		K_{zt}=1.0
	WIND DIRECTIONALITY FACTOR		K_d =0.95
SEISMIC LOAD DATA	SITE CLASS		B
	SEISMIC USER GROUP		SUG - I
	IMPORTANCE FACTOR	(I)	1
	SCALING FACTOR	(Q)	2/3
	MAXIMUM DESIGN LIQUID LEVEL	(H)	15.860 m
	CO-EFFICIENT TO ADJUST THE SPECTRAL ACCELERATION FROM 5% TO 0.5% DAMPING	(K)	1.50
	SEISMIC CO-EFFICIENTS		S_s= 0.1g, S_1= 0.02g, F_a = 1.0, F_v= 1.0 S_{ds} = 0.067g, S_{d1}= 0.013g, T = 4 sec.
OTHER DATA	ROOF LIFE LOAD		1.2 kPa
	FOUNDATION TYPE (RING WALL/SLAB)		RING WALL
	PWHT REQUIRED(YES/NO)- IF YES,SPECIFY REASON FOR PWHT		NO
	REASON FOR PWHT(PROCESS REASON/PER API CODE)		AS PER API 650
	RADIOGRAPHY		AS PER CL 8.1 - API 650
	AGITATOR RAKE REQUIRED	(YES/NO)	NO
		EQUIPMENT TYPE (AGITATOR/RAKE)	N/A
		AGITATOR/RAKE SUPPORTED BY TANK (YES/NO)	N/A
	INSULATION	SHELL	N/A
		ROOF	N/A
	LINER REQUIRED	(YES/NO)	NO
		LINER MATERIAL	N/A
		LINER THICKNESS (ROOF/SHELL/BOT)	N/A
	PAINTING	INTERNAL	YES (REFER NOTE 14) A02
		EXTERNAL	YES
	PIPE SUPPORT		YES
	SPIRAL STAIRWAY		NO
	ROOF CIRCUMFERENTIAL HANDRAIL		YES
	WIND GIRDER		AS REQUIRED
	CATHODIC PROTECTION		NO
	LEAK DETECTION		NO
	INTERNAL BAFFLES		NO
	EARTHING LUGS		YES





PLAN
(SCALE - 1: 50) A02

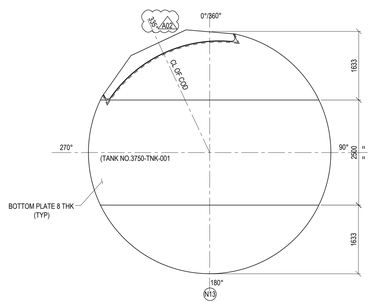
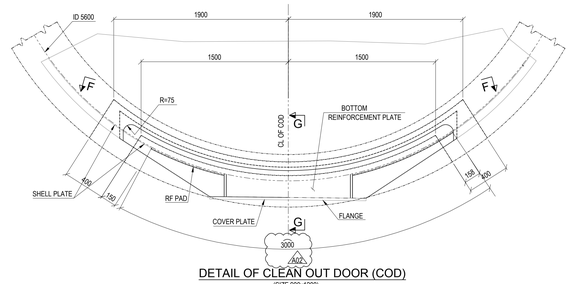
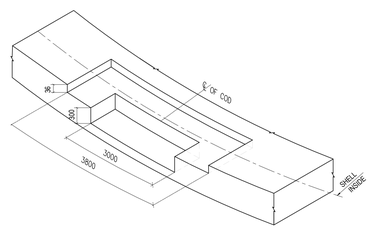


PLATE ARRANGEMENT OF BOTTOM PLATE
(SCALE - 1: 50)



DETAIL OF CLEAN OUT DOOR (COD)
(SIZE: 900x1200)
SCALE- 1:20



3D VIEW OF FOUNDATION NOTCH
(FOR COD)

LEGEND:

ID - INNER DIAMETER
OD - OUTER DIAMETER
COD - CLEAN OUT DOOR
LG - LONG
FGL - FINISHED GROUND LEVEL.
BOT - BOTTOM OF TANK
TOT - TOP OF TANK
RL - REDUCED LEVEL
RF - REINFORCEMENT
TOC - TOP OF CONCRETE
GST - GUSSET
PL - PLATE
STIFF - STIFENER
THK - THICKNESS
CLT - CLEAT
UNO - UNLESS NOTED OTHERWISE
BOT - BOTTOM OF TANK
TOT - TOP OF TANK
NOZ - NOZZLE
DN - DIAMETER NOMINAL
SCH - SCHEDULE
ELVN - ELEVATION
PROJ - PROJECTION
MM - MILLIMETER
TYP - TYPICAL
GA - GENERAL ARRANGEMENT
DRG - DRAWING
CL - CENTER LINE
TOS - TOP OF STEEL
WP - WORKING POINT
N/S - NEAR SIDE
F/S - FAR SIDE
DIA - DIAMETER
C/F - CUT FORM
TBC - TO BE CONFIRMED
TOG - TOP OF GRATING
N/A - NOT APPLICABLE
ATM - ATMOSPHERIC
MIN - MINIMUM

UNDER PREPARATION

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36,
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
4. BOLT HOLES IN NOZZLE FLANGES & MANHOLES SHALL STRADDLE VERTICAL CENTER LINES ON SHELL & NORTH SOUTH CENTERLINES ON ROOF.
5. NOZZLE PROJECTION ON SHELL SIDE REFERS FROM EXTERNAL SURFACE OF SHELL PLATE TO FACE OF FLANGE. NOZZLE PROJECTION ON ROOF SIDE REFERS FROM TOP OF ROOF PLATE TO FACE OF FLANGE.
6. VERTICAL SHELL JOINTS SHOULD PREFERABLY BE OFFSET BY 1000 MM.
7. CURB ANGLE JOINTS SHALL STAGGER ATLEAST 300 MM WITH VERTICAL SHELL PLATE WELD.
8. REINFORCEMENT PADS TO HAVE TELLTALE HOLES OF 6 MM DIA AND SHALL BE TESTED AT 100 kPa PRESSURE WITH SOAP SOLUTION AND THEN FILLED WITH HARD GREASE.
9. FLANGE (COD & MANHOLE) GASKET SEATING SURFACE SHALL BE PROVIDED WITH 0.0032 TO 0.0063 mm (3.2 TO 6.3 µm) ROUGHNESS FINISH.
10. EARTHING LUG DETAIL & LOCATION SHALL BE CONFIRMED IN SEPARATE DRAWING.
11. SUITABLE NOTCHES AS PER API 650 TO BE PROVIDED IN THE FOUNDATION/GROUT FOR THE COD.
12. MINIMUM THICKNESS OF RF PAD, FLANGE AND COVER PLATES FOR NOZZLES ARE PROVIDED BASED ON HYDROSTATIC LOAD AND FINAL SIZE SHALL BE CONFIRMED AFTER NOZZLE DESIGN IN SEPERATE DOCUMENT.
13. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3750-EG -ST - GAD - 1002 & 1003.
14. PAINTING SHALL CONFORM TO MAADEN SPECIFICATION NUMBER MD-101-SMEM-EG-ME-SPC-0007 AND SEPARATE PAINTING DETAILS & APPLICATION PROCEDURE WILL BE SUBMITTED FOR APPROVAL. A02

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS.	A02	23/04/2020	REVISED AS PER EMPLOYER'S.	AKDS	PRI	HKA	AG	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA			COMMENTS ON 25/03/2020 & REVISED AS MKD. A02					
MD-216-3750-EG-ME-GAD-1001	CIL PROCESS WATER TANK 3750-TNK-001	A01	29/02/2020	ISSUED FOR REFERENCE	SJH	PRI	HKA	AG	

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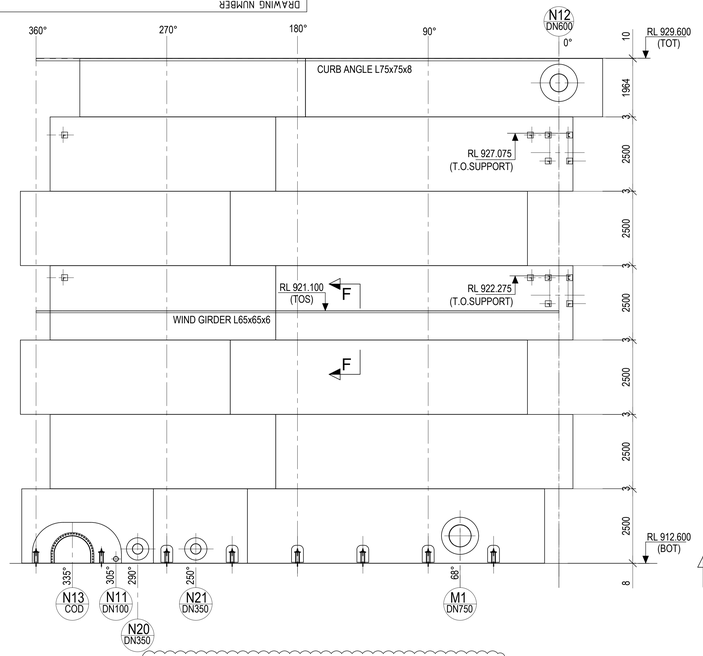
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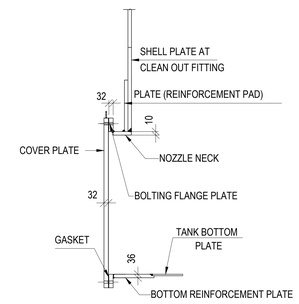
LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_450_S_G_26150_001			1
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707__GAU01__AG03__MTL01__37502		OU1006734-92	1
DRAWN	SJH	22/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	PRI	28/02/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	HKA	27/02/2020	TAILING & CIL PROCESS WATER STORAGE
APPROVED	AG	29/02/2020	PLAN, ELVN, DETAIL OF TANK 3750-TNK-001
EMPLOYER			DRAWING NUMBER: MD - 216 - 3750 - EG - ST - GAD - 1001
SCALE 1:100,20		A0	REV A02

DRAWING NUMBER MD-216-3750-EG-ST-GAD-1002 REV A02



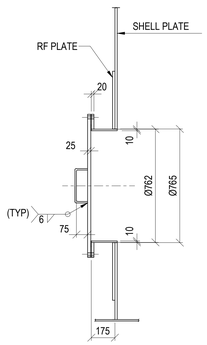
ELEVATION OF DEVELOPED SHELL PLATE (FOR TANK NO 3750-TNK-001) A02

(SCALE - 1: 100)



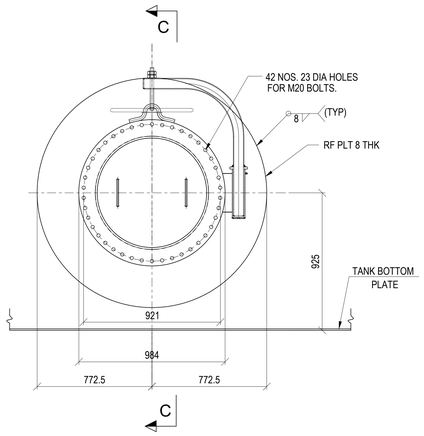
SECTION B-B

SCALE- 1:20



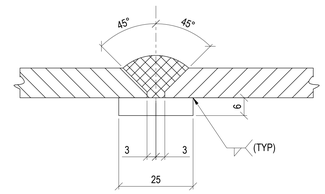
SECTION C-C

SCALE- 1:20

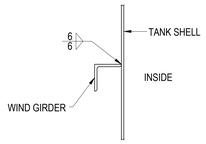


TYP. DETAIL OF SHELL DAVIT MANHOLE

SCALE- 1:20

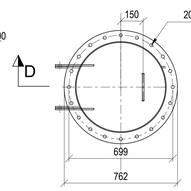


TYP.BOTTOM PLATE JOINT DETAIL



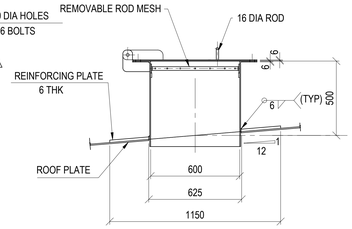
SECTION F-F

SCALE- 1:10



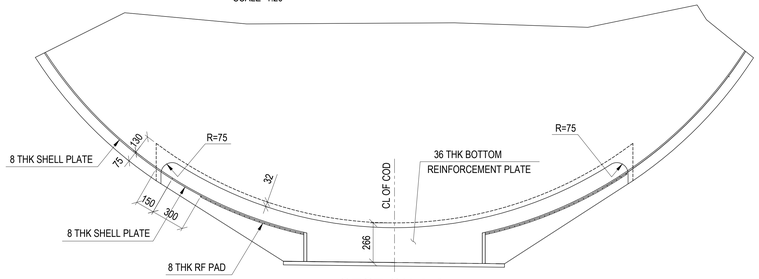
TYP. DETAIL OF ROOF MANHOLE (600 DN)

SCALE- 1:20



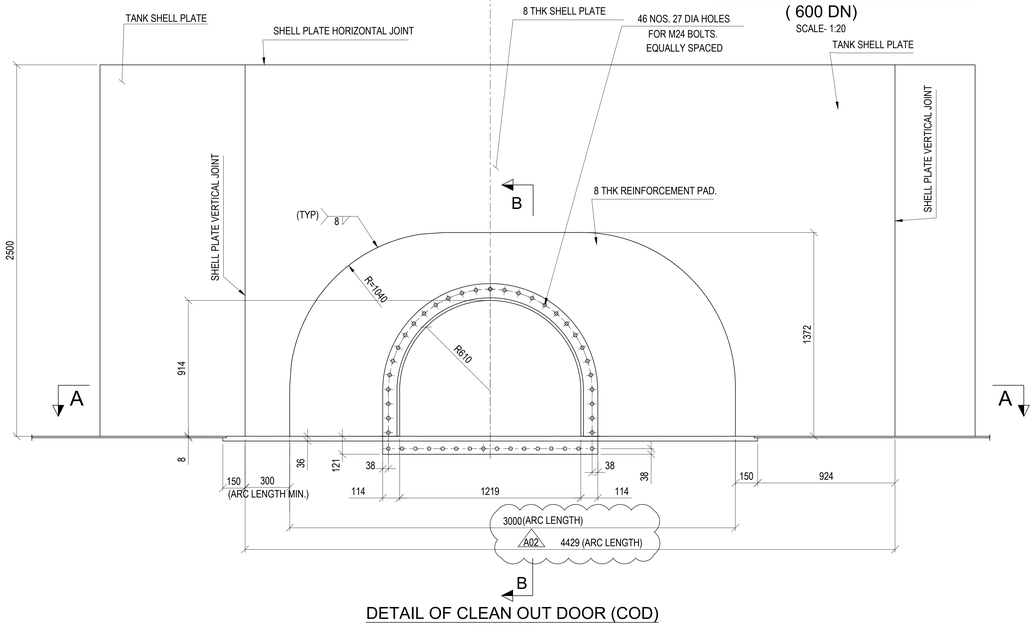
SECTION D-D

SCALE- 1:20



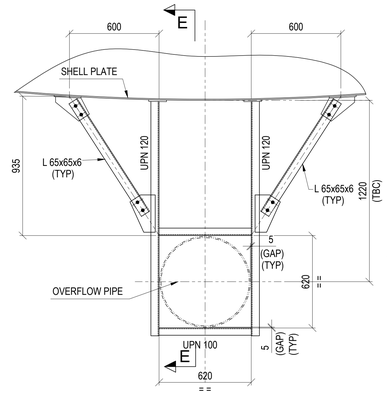
SECTION A-A

SCALE- 1:20



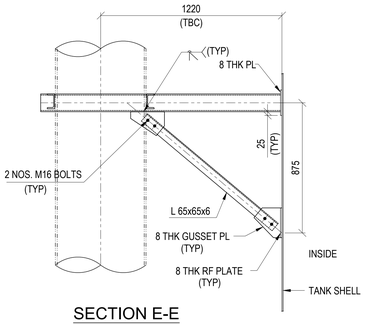
DETAIL OF CLEAN OUT DOOR (COD)

(SIZE 900x1200)
SCALE- 1:20



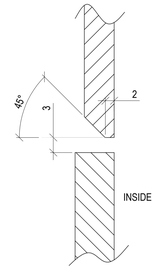
PLAN FOR OVERFLOW SUPPORT.

SCALE- 1:20



SECTION E-E

SCALE- 1:20



TYP.SHELL PLATE JOINT DETAIL

UNDER PREPARATION

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
3. FOR ALL OTHER NOTES REFER DRAWING MD - 216 - 3750 -EG -ST - GAD - 1001.
4. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
5. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3750 -EG -ST - GAD -1001 & 1003.
6. FOR LEGEND LIST REFER DRAWING MD - 216 - 3750 -EG -ST - GAD - 1001.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS.	A02	23/04/2020	REVISED AS PER EMPLOYER'S					
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA			COMMENTS ON 25/03/2020 & REVISED AS MKD. A02	AKDS	PRI	HKA	AG	
MD-216-3750-EG-ME-GAD-1001	CIL PROCESS WATER TANK 3750-TNK-001	A01	29/02/2020	ISSUED FOR REFERENCE	SJH	PRI	HKA	AG	

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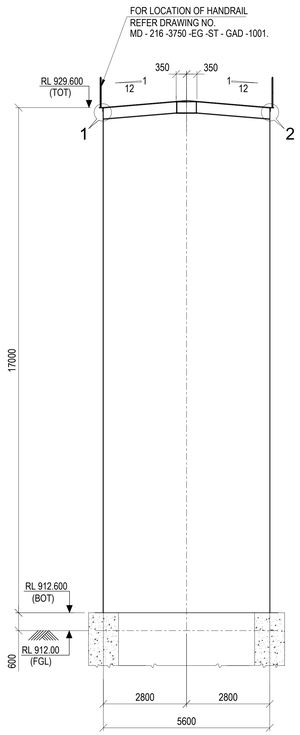
OUTOTEC DRAWING NUMBER: 900707 __GAU01 __AG03__MTL01 __37503 — DOCUMENT ID: OU100673493 — REVISION: 1

DRAWN	SJH	22/02/2020
CHECKED	PRI	28/02/2020
ENGINEER	HKA	27/02/2020
APPROVED	AG	29/02/2020
EMPLOYER		
SCALE	1:100,20,10	A1

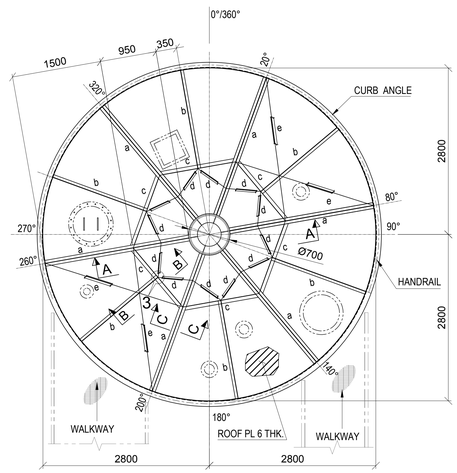
MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TAILING & CIL PROCESS WATER STORAGE
SHELL PLT DETAIL OF TANK 3750-TNK-001

DRAWING NUMBER: MD - 216 - 3750 - EG - ST - GAD - 1002 — REV: A02

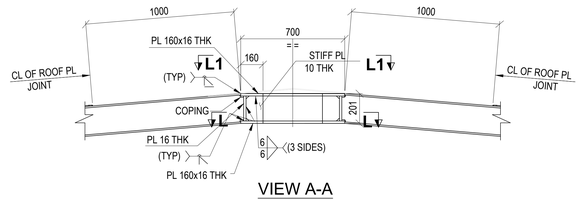
DRAWING NUMBER MD - 216 - 3750 - EG - ST - GAD - 1003 | REV A02



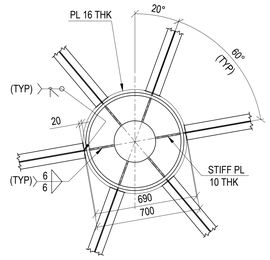
ELEVATION SHOWING RAFTER
(SCALE 1:100)



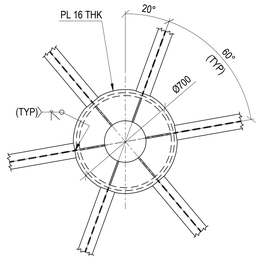
PLAN AT ROOF LEVEL
(SCALE 1:50) A02



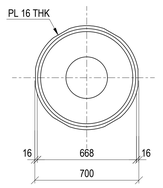
VIEW A-A
(TYP. CONNECTION OF MAIN RAFTER WITH CENTRAL DRUM)
(SCALE 1:20)



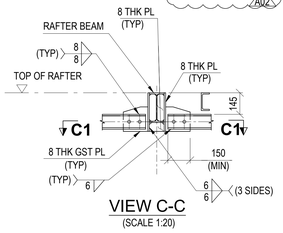
SECTION L-L
(SCALE 1:20) A02



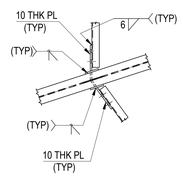
SECTION L1-L1
(SCALE 1:20) A02



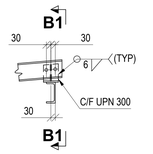
TOP & BOTTOM PLATE
(SCALE 1:20)



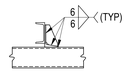
VIEW C-C
(SCALE 1:20)



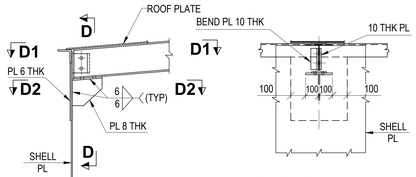
SECTION C1-C1
(SCALE 1:20)



VIEW B-B
(SCALE 1:20)

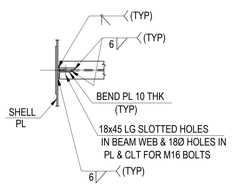


SECTION B1-B1
(SCALE 1:20)

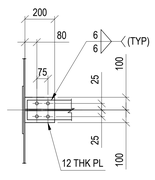


DETAIL-1
(SCALE 1:20)

SECTION D-D
(SCALE 1:20)



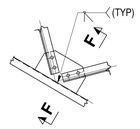
SECTION D1-D1
(SCALE 1:20)

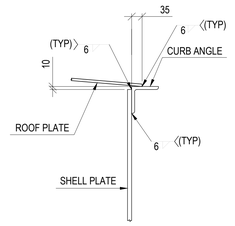


SECTION D2-D2
(SCALE 1:20)



SECTION F-F
(SCALE 1:20)





DETAIL- 2
(TYP CONNECTION OF SHELL PL ROOF PL & CURB ANGLE)
(SCALE 1:10)

SCHEDULE OF MEMBERS

MEMBER MARKED	MEMBER	SECTION	FACTORED AXIAL FORCE(kN)	FACTORED SHEAR FORCE(kN)	REMARKS
a	IPE200	I	40	30	MATERIAL GRADE :S275 JR
b	UPN120	[	25	15	
c	UPN 120	[	30	20	
d	L 50X50X6	⌐	15	-	
e	L 65X65X6	⌐	20	-	

FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED. MINIMUM 6 MM FILLET WELD OF 150 MM LENGTH SHALL BE PROVIDED. ALL GUSSET PLATE ARE 8 MM THK (UNO).

UNDER PREPARATION

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
3. FOR ALL OTHER NOTES REFER DRAWING MD - 216 - 3750 -EG -ST - GAD - 1001.
4. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
5. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3750 -EG -ST - GAD - 1001 & 1002.
6. FOR LEGEND LIST REFER DRAWING MD - 216 - 3750 -EG -ST - GAD - 1001.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS.	A02	23/04/2020	REVISED AS PER EMPLOYER'S					
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA			COMMENTS ON 25/03/2020 & REVISED AS MKD. A02	AKDS	PRI	HKA	AG	
MD-216-3750-EG-ME-GAD-1001	CIL PROCESS WATER TANK 3750-TNK-001	A01	29/02/2020	ISSUED FOR REFERENCE	SJH	PRI	HKA	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER		REVISION
O19045_450_S_G_26150_003		1
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707 __GAU01 __AG03__MTL01 __37504	OU100673494	1

DRAWN	SJH	22/02/2020
CHECKED	PRI	28/02/2020
ENGINEER	HKA	27/02/2020
APPROVED	AG	29/02/2020
EMPLOYER		
SCALE	1:100,50,20	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TAILING & CIL PROCESS WATER STORAGE
ROOF SYSTEM OF TANK 3750-TNK-001

DRAWING NUMBER: MD - 216 - 3750 - EG - ST - GAD - 1003 | REV: A02

ATTACHMENT 3 STAAD OUTPUT

PROPRIETARY & CONFIDENTIAL

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    MAY 21, 2020                   *
*           Time=    21:58:31                       *
*                                                   *
*      USER ID: LNT ECC DIV                         *
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    1. STAAD SPACE
INPUT FILE: TANK FDN (3750).STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 10.05.2020
    4. JOB NAME MANSOURAH MASSARAH GOLD PROJECT
    5. JOB CLIENT MA'ADEN
    6. ENGINEER NAME SOR
    7. CHECKER NAME SJM
    8. APPROVED NAME AG
    9. JOB REV 0
   10. END JOB INFORMATION
   11. INPUT WIDTH 79
   12. UNIT MMS KN
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172. ELEMENT PROPERTY
173. 6436 TO 6507 6532 TO 6555 THICKNESS 600
174. 6782 TO 6785 6799 TO 6802 6921 TO 6988 7212 TO 7215 7218 7222 7223 7226 7231 -
175. 7232 TO 7234 7248 TO 7251 7261 7262 7269 7276 7320 TO 7332 7334 TO 7336 -
176. 7337 THICKNESS 500
177. DEFINE MATERIAL START
178. ISOTROPIC CONCRETE
179. E 21.7185
180. POISSON 0.17
181. DENSITY 2.35616E-008
182. ALPHA 1E-005
183. DAMP 0.05
184. TYPE CONCRETE
185. STRENGTH FCU 0.027579
186. END DEFINE MATERIAL
187. CONSTANTS
188. MATERIAL CONCRETE ALL
189. START GROUP DEFINITION
190. ELEMENT
191. _RAFT 6782 TO 6785 6799 TO 6802 6921 TO 6988 7212 TO 7215 7218 7222 7223 7226 -
192. 7231 TO 7234 7248 TO 7251 7261 7262 7269 7276 7320 TO 7332 7334 TO 7337
193. _WALL 6436 TO 6507 6532 TO 6555
194. END GROUP DEFINITION
195. UNIT METER KN
196. SUPPORTS
197. 6782 TO 6785 6799 TO 6802 6921 TO 6988 7212 TO 7215 7218 7222 7223 7226 7231 -
198. 7232 TO 7234 7248 TO 7251 7261 7262 7269 7276 7320 TO 7332 7334 TO 7336 -
199. 7337 PLATE MAT DIRECT Y SUBGRADE 85500
200. ********************************************************************************
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202. SELFWEIGHT Y -1
203. JOINT LOAD
204. 3038 TO 3061 FY -11.25
205. ELEMENT LOAD
206. 6782 TO 6785 6799 TO 6802 6921 TO 6988 PR GY -32
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241. 6444 TRAP JT -8 -8 -12 -12
242. 6445 TRAP JT -8 -8 -12 -12
243. 6446 TRAP JT -8 -8 -12 -12
244. 6447 TRAP JT -8 -8 -12 -12
245. 6448 TRAP JT -8 -8 -12 -12
246. 6449 TRAP JT -8 -8 -12 -12
247. 6450 TRAP JT -8 -8 -12 -12
248. 6451 TRAP JT -8 -8 -12 -12
249. 6452 TRAP JT -8 -8 -12 -12
250. 6453 TRAP JT -8 -8 -12 -12
251. 6454 TRAP JT -8 -8 -12 -12
252. 6455 TRAP JT -8 -8 -12 -12
253. 6456 TRAP JT -8 -8 -12 -12
254. 6457 TRAP JT -8 -8 -12 -12
255. 6458 TRAP JT -8 -8 -12 -12
256. 6459 TRAP JT -8 -8 -12 -12
257. 6460 TRAP JT -4 -4 -8 -8
258. 6461 TRAP JT -4 -4 -8 -8
259. 6462 TRAP JT -4 -4 -8 -8
260. 6463 TRAP JT -4 -4 -8 -8
261. 6464 TRAP JT -4 -4 -8 -8
262. 6465 TRAP JT -4 -4 -8 -8
```



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263. 6466 TRAP JT -4 -4 -8 -8
264. 6467 TRAP JT -4 -4 -8 -8
265. 6468 TRAP JT -4 -4 -8 -8
266. 6469 TRAP JT -4 -4 -8 -8
267. 6470 TRAP JT -4 -4 -8 -8
268. 6471 TRAP JT -4 -4 -8 -8
269. 6472 TRAP JT -4 -4 -8 -8
270. 6473 TRAP JT -4 -4 -8 -8
271. 6474 TRAP JT -4 -4 -8 -8
272. 6475 TRAP JT -4 -4 -8 -8
273. 6476 TRAP JT -4 -4 -8 -8
274. 6477 TRAP JT -4 -4 -8 -8
275. 6478 TRAP JT -4 -4 -8 -8
276. 6479 TRAP JT -4 -4 -8 -8
277. 6480 TRAP JT -4 -4 -8 -8
278. 6481 TRAP JT -4 -4 -8 -8
279. 6482 TRAP JT -4 -4 -8 -8
280. 6483 TRAP JT -4 -4 -8 -8
281. 6484 TRAP JT 0 0 -4 -4
282. 6485 TRAP JT 0 0 -4 -4
283. 6486 TRAP JT 0 0 -4 -4
284. 6487 TRAP JT 0 0 -4 -4
285. 6488 TRAP JT 0 0 -4 -4
286. 6489 TRAP JT 0 0 -4 -4
287. 6490 TRAP JT 0 0 -4 -4
288. 6491 TRAP JT 0 0 -4 -4
289. 6492 TRAP JT 0 0 -4 -4
290. 6493 TRAP JT 0 0 -4 -4
291. 6494 TRAP JT 0 0 -4 -4
292. 6495 TRAP JT 0 0 -4 -4
293. 6496 TRAP JT 0 0 -4 -4
294. 6497 TRAP JT 0 0 -4 -4
295. 6498 TRAP JT 0 0 -4 -4
296. 6499 TRAP JT 0 0 -4 -4
297. 6500 TRAP JT 0 0 -4 -4
298. 6501 TRAP JT 0 0 -4 -4
299. 6502 TRAP JT 0 0 -4 -4
300. 6503 TRAP JT 0 0 -4 -4
301. 6504 TRAP JT 0 0 -4 -4
302. 6505 TRAP JT 0 0 -4 -4
303. 6506 TRAP JT 0 0 -4 -4
304. 6507 TRAP JT 0 0 -4 -4
305. ******************************************************************************
306. LOAD 2 LOADTYPE FLUIDS  TITLE OPERATING LIQUID LOAD ( F1 )
307. ELEMENT LOAD
308. 6782 TO 6785 6799 TO 6802 6921 TO 6988 PR GY -155.5
309. 6436 TO 6507 6532 TO 6555 PR -77.75
310. ******************************************************************************
311. LOAD 3 LOADTYPE FLUIDS  TITLE TEST LOAD ( F2 )
312. ELEMENT LOAD
313. 6782 TO 6785 6799 TO 6802 6921 TO 6988 PR GY -166.8
314. 6436 TO 6507 6532 TO 6555 PR -83.4
315. ******************************************************************************
316. LOAD 4 LOADTYPE WIND  TITLE WIND LOAD ( W )
317. JOINT LOAD
318. 3038 TO 3061 FZ -2.25
```

```
319. 3042 FY 0
320. 3043 FY 4.3
321. 3044 FY 8.3
322. 3045 FY 11.8
323. 3046 FY 14.4
324. 3047 FY 16.1
325. 3048 FY 16.7
326. 3049 FY 16.1
327. 3050 FY 14.4
328. 3051 FY 11.8
329. 3052 FY 8.3
330. 3053 FY 4.3
331. 3054 FY 0
332. 3055 FY -4.3
333. 3056 FY -8.3
334. 3057 FY -11.8
335. 3058 FY -14.4
336. 3059 FY -16.1
337. 3060 FY -16.7
338. 3061 FY -16.1
339. 3038 FY -11.8
340. 3039 FY -14.4
341. 3040 FY -8.3
342. 3041 FY -4.3
343. ******************************************************************************
344. LOAD 5 LOADTYPE SEISMIC  TITLE SEISMIC LOAD ( E )
345. JOINT LOAD
346. 3038 TO 3061 FZ -2.708
347. 3042 FY 0
348. 3043 FY 3.8
349. 3044 FY 7.4
350. 3045 FY 10.5
351. 3046 FY 12.9
352. 3047 FY 14.3
353. 3048 FY 14.8
354. 3049 FY 14.3
355. 3050 FY 12.9
356. 3051 FY 10.5
357. 3052 FY 7.4
358. 3053 FY 3.8
359. 3054 FY 0
360. 3055 FY -3.8
361. 3056 FY -7.4
362. 3057 FY -10.5
363. 3058 FY -12.9
364. 3059 FY -14.3
365. 3060 FY -14.8
366. 3061 FY -14.3
367. 3038 FY -10.5
368. 3039 FY -12.9
369. 3040 FY -7.4
370. 3041 FY -3.8
371. ******************************************************************************
372. LOAD 6 LOADTYPE FLUIDS  TITLE LIVE LOAD ( L )
373. JOINT LOAD
374. 3038 TO 3061 FY -3.5
```



```
375. ******************************************************************************
376. * LOAD COMBINATIONS
377. LOAD COMB 201 1.4 ( D + F1 )
378. 1 1.4 2 1.4
379. LOAD COMB 202 1.4 ( D + F1 ) + 1.7 L
380. 1 1.4 2 1.4 6 1.7
381. LOAD COMB 203 0.9 D + 1.6 W
382. 1 0.9 4 1.6
383. LOAD COMB 204 1.2 ( D + F1 ) + 1.6 W + L
384. 1 1.2 2 1.2 4 1.6 6 1.0
385. LOAD COMB 205 1.2 ( D + F1 ) + E + L
386. 1 1.2 2 1.2 5 1.0 6 1.0
387. LOAD COMB 206 0.9 ( D + F1 ) + E
388. 1 0.9 2 0.9 5 1.0
389. LOAD COMB 207 1.4 ( D + F2 )
390. 1 1.4 3 1.4
391. LOAD COMB 208 1.4 ( D + F2 ) + 1.7 L
392. 1 1.4 3 1.4 6 1.7
393. LOAD COMB 209 1.2 ( D + F2 ) + 1.6 W + L
394. 1 1.2 3 1.2 4 1.6 6 1.0
395. LOAD COMB 210 1.2 ( D + F2 ) + E + L
396. 1 1.2 3 1.2 5 1.0 6 1.0
397. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 209 NUMBER OF MEMBERS 0
NUMBER OF PLATES 209 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 113

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 203/ 29/ 138 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 915
TOTAL LOAD COMBINATION CASES = 10 SO FAR.
SIZE OF STIFFNESS MATRIX = 127 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 14.5/ 146355.7 MB

```
398. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 21,2020 TIME= 21:58:37 ****

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* *
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